A-A-50967A
February 28, 1990
SUPERSEDING
A-A-50967
April 18, 1984

COMMERCIAL ITEM DESCRIPTION

GRINDING MACHINE, BENCH, HAND POWERED

The General Services Administration has authorized the use of this commercial item description.

Abstract. This commercial item description covers hand powered, bench mounting, grinding machines with a six inch diameter grinding wheel used for sharpening tools. The grinding machine is intended for those applications where it is impractical to use power driven type grinding machines.

Classification. The machine shall be of the following types. The type shall be as specified.

Type I - Heavy duty - Rotation of the grinding wheel is accomplished through the use of a gear mechanism with a ratio of 13 to 1 or more.

Type II - Light duty - Rotation of the grinding wheel is accomplished through the use of a gear mechanism with a ratio of 9 to 1 or more.

Salient characteristics.

1. Design. The hand powered bench grinding machine shall be new and of the manufacturer's commercial design in effect on the date of the solicitation. The machine shall meet the requirements specified herein and conform to current industrial practices. The operation of the machine shall be smooth. There shall be no wheel slippage during the grinding operations. The machine shall be tested by grinding an edge on a sample workpiece of 1018 steel not less than 7 gauge x 1 inch x 6 inches. The machine shall demonstrate compliance with the salient characteristics during the test.

2. Bushings. Grinding machines shall have not less than four bushing supports. Two of the bushings shall support the grinding wheel shaft and two shall support the hand crank shaft. The bushings shall be of the cast iron type.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd., Memphis, TN 38114-5051.

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3. Gear case. The grinding machine gear case shall be the same as that provided on the manufacturers' standard product and shall have the proper size and strength to support the grinding machine components. All encased gears shall operate smoothly and without excessive noise.

4. Gears. All gears shall be constructed from a material suitable for the intended purpose. All drive gears shall have the hardness, surface finish, and toughness that will enable the gear train to transmit full rated torque of the drive without gear damage, failure, or premature deterioration and wear.

5. Hand crank. A hand crank and handle shall be provided to impart rotary motion to the grinding wheel. The hand crank and gears shall be tested by clamping the driving gear shaft so that it remains stationary. The type I machine shall have a force of 120 inch-pounds applied to the crank handle in the clockwise direction of rotation using a spring balance. The movement of the crank handle from the stationary position shall be marked. A force of 120 inch-pounds shall be applied to the crank handle in the counterclockwise direction of rotation using a spring balance and the movement shall be marked as above. The difference in the clockwise and counterclockwise movement of the crank handle shall not exceed 1/4 inch. The type II machine shall have an identical test to the type I machine with the exception of using 60 inch-pounds of force applied to the handle. The difference in the clockwise and counterclockwise movement of the crank handle shall not exceed 1/4 inch. Both types of machines shall be further tested by applying a force of 100 inch-pounds to the end of the crank handle in a direction tending to pull the crank away from the gear case. The crank handle shall not break or show permanent deformation.

6. Tool rest. The machine shall be provided with a tool rest. The tool rest shall be adjustable to various heights and positions to accommodate edged hand tools.

7. Grinding wheel. Grinding wheel dimensions shall be six inches in diameter, one inch wide, and be provided with a 3/8 inch diameter hole. The grinding wheel dimensions are applicable to both type I and II machines. The wheel shall have aluminum oxide abrasives, vetrified bond, medium grain, medium grade of hardness, square faced and straight sides, and shall meet the requirements of ANSI B74.13.

8. Mounting flanges. Mounting flanges shall be provided to prevent slippage of the grinding wheel during operation.

9. Screw clamp. A screw clamp shall be furnished with type I machines and shall accommodate bench tops one to three inches in thickness. A screw clamp shall be furnished with type II machines which shall accommodate bench tops 3/4 to 1-1/2 inches in thickness. The screw clamp shall be tested after securely attaching the grinder to a bench top. The type I machines shall be subjected to a force or torsional moment of 120 inch-pounds using a torque wrench, spring balance, or other load measuring device. The type II machines shall be subjected to a force or torsional moment of 60 inch-pounds using either of the same devices. The force shall be applied at right angles to the longitudinal axis of the screw. No parts of the adjusting screw or clamp shall bend, break, or show any visual indications of weakness.

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10. Lubrication. A suitable means of lubrication shall be provided with the machine that lubricates all bushings and gears. Hinge lid oil caps shall be provided on all external oil holes to prevent grit or dust from entering.

11. Paint. External surfaces shall be painted in accordance with the manufacturer's standard practice and color.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this commercial item description are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request shall be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing suppliers' employees to undue hazards from the recovered materials.

Environmental protection. During the manufacture of the machine, materials hazardous to the ecological system as prohibited by the Federal, state, or local statutes in effect on the date of the contract shall not be used or emitted.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection, examination, and test requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections, examinations, or tests set forth in this description where such inspections, examinations, and tests are deemed necessary to assure supplies and services conform to prescribed requirements.

Responsibility for compliance. All items shall meet all requirements of salient characteristics paragraphs. The inspection set forth in this description shall become a part of the contractors overall inspection system or quality program. The absence of any inspection requirements in the description shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance

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comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

Sampling inspection. As a minimum sampling inspection, the supplier shall randomly select the number of samples in table I that is applicable to the quantity of items being purchased. Each sample item shall be inspected to determine conformance with the examination, the test, and the inspection. Should any of the sample items fail to meet either the examination or inspection below, the entire lot shall be considered defective and unsuitable for acceptance. A lot is defined as all items presented at one time for acceptance.

TABLE I. Sampling plan.

Quantity of items being purchased	Sample size	Quantity of items being purchased	Sample size
2 to 8	2	151 to 280	32
9 to 15	3	281 to 500	50
16 to 25	5	501 to 1,200	80
26 to 50	8	1,201 to 3,200	125
51 to 90	13	3,201 to 10,000	200
91 to 150	20		

Examination. Each grinding machine submitted for sample inspection shall be visually examined to determine conformance with all requirements of this description.

Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be as specified in the contract or order.

Cross-reference of classification changes. Classification changes from the previous issue of this commercial item description are as listed in table II.

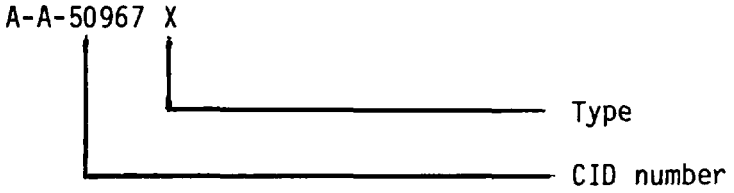
Table II. Cross-reference of hand powered bench grinding machines.

A-A-50967	A-A-50967A
Type I, size 6	(Deleted)
Type I, size 7	(Deleted)
Type II, size 4	(Deleted)
Type II, size 5	(Deleted)
Type II, size 6	(Deleted)
Type II, size 7	(Deleted)
	Type I
	Type II

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NOTE: This commercial item description lists no sizes. Both the type I and type II machines have only a six inch diameter grinding wheel.

CID based part identification number. The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor. Where applicable Arabic numbers shall be substituted for Roman numerals.



Notes.

Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number, and date of this commercial item description.

b. Type and quantity required.

c. Other requirements.

ANSI B74.13 is available from the American National Standards Institute, Attn: Sales Department, 1430 Broadway, New York, NY 10018-3363 (DOD ADOPTED).

MILITARY INTEREST:

Custodians

Air Force - 99

Review Activities

Air Force - 84

User Activities

DLA-GS

CIVIL AGENCY COORDINATION ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

DLA - IP

Project 3415-0185

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NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER A-A-50967A	2. DOCUMENT DATE (YYMMDD) 90-02-28

3. DOCUMENT TITLE
GRINDING MACHINE, BENCH, HAND POWERED

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON
c. ADDRESS *(Include Zip Code)*	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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